

Neuralgic Amyotrophy (Parsonage–Turner Syndrome): A Comprehensive Disease Characterization

Summary

Neuralgic Amyotrophy (NA), also known as **Parsonage–Turner syndrome (PTS)** or **brachial plexus neuritis**, is a distinct, painful, multifocal peripheral neuropathy that predominantly targets the brachial plexus and its terminal branches. Its clinical hallmark is an abrupt onset of severe, continuous neuropathic pain in the shoulder and upper arm, which after roughly four weeks gives way to patchy flaccid paresis, muscle atrophy, and sensory disturbance. NA exists in two forms: a common **idiopathic form (INA)** and a rarer **autosomal-dominant hereditary form (HNA)**, the latter caused in about 55% of families by mutations in the **SEPT9 (SEPTIN9)** gene on chromosome 17q25. Once regarded as a rare and benign self-limited condition, NA is now recognized as **common (incidence ~1 per 1,000 per year)**, **profoundly underdiagnosed**, and **prognostically guarded**, with a majority of patients left with residual functional limitations.

The pathophysiology reflects a **three-factor model**: an underlying genetic predisposition, a mechanical susceptibility of the brachial plexus (likely involving the epineurial blood–nerve barrier), and an **immune/autoimmune trigger** for individual attacks. Triggers include infection (notably **hepatitis E virus, HEV**), vaccination (including COVID-19 mRNA and viral-vector vaccines), surgery, trauma, childbirth, and strenuous exercise. The convergence of these factors produces a focal inflammatory neuritis whose imaging signature is the **pathognomonic hourglass-like nerve constriction (HGC)**, detectable by high-resolution ultrasound and MR neurography in roughly three-quarters of acutely imaged patients. Direct histopathologic evidence of multifocal mononuclear inflammatory infiltrates on nerve biopsy supports the immune basis, and a naturally occurring inflammatory brachial plexus neuritis in cats provides a comparative-species analogue.

Management centers on **early high-dose corticosteroids** for pain control in the acute phase, **motor-relearning multidisciplinary rehabilitation** (rather than strength training) for the chronic phase, and **microsurgical neurolysis** of hourglass constrictions in selected patients

who fail to recover. Diagnosis remains fundamentally clinical and one of exclusion, supported by electrodiagnostics, modern nerve imaging, HEV serology (particularly when the phrenic nerve/diaphragm is involved), and SEPT9 genetic testing in hereditary or pediatric cases. This report synthesizes eight confirmed findings drawn from 47 reviewed papers across all 15 requested disease-characteristic domains.

1. Disease Information

Overview. Neuralgic Amyotrophy is an uncommon-to-underrecognized peripheral nervous-system disorder characterized by episodes ("attacks") of extreme neuropathic pain followed by rapid, multifocal, patchy weakness and atrophy predominantly in the upper limbs. In its classic presentation it is a brachial plexopathy or multifocal motor-predominant neuropathy with a monophasic course, though sensory involvement is common and extra-plexus nerves are frequently affected.

Key identifiers (as commonly catalogued): - **MONDO:** MONDO:0100053 (neuralgic amyotrophy) — suggested; confirm in current release - **OMIM:** 162100 (Neuralgic amyotrophy / hereditary brachial plexus neuropathy, HNA) - **Orphanet:** ORPHA:2901 (Neuralgic amyotrophy) - **ICD-10:** G54.5 (Neuralgic amyotrophy) - **ICD-11:** 8B93 / brachial plexus disorder region (neuralgic amyotrophy) - **MeSH:** D020968 (Brachial Plexus Neuritis) - **Gene (HNA):** SEPT9 / SEPTIN9, HGNC:7323, chromosome 17q25

Synonyms and alternative names: Parsonage–Turner syndrome; brachial plexus neuritis; brachial neuritis; brachial neuralgic amyotrophy; idiopathic brachial plexopathy; acute brachial neuropathy; hereditary brachial plexus neuropathy (HBPN, hereditary form); shoulder-girdle syndrome. Parsonage and Turner unified previously separate descriptions under the term "neuralgic amyotrophy" in 1948, highlighting the combination of neuropathic pain and muscular atrophy ([PMID: 39070412](#)).

Source of information. The knowledge in this report is derived from **aggregated disease-level resources** — cohort studies, case-control studies, clinical reviews, and case reports/series — rather than individual EHR extraction. The largest supporting cohort comprised 246 patients ([PMID: 16371410](#)).

2. Etiology

Disease causal factors. NA is a **complex disease** arising from the interaction of genetic predisposition, mechanical vulnerability, and immune-mediated triggering. There is no single cause; rather, the current, well-supported model is a **triad** ([PMID: 21556032](#)):

"The disease is thought to result from an underlying genetic predisposition, a susceptibility to mechanical injury of the brachial plexus (possibly representing disturbance of the epineurial blood-nerve barrier), and an immune or autoimmune trigger for the attacks."

Genetic risk factors. In the hereditary form (HNA), **SEPT9** point mutations and duplications on 17q25 are causal in ~55% of families (see Section 4). SEPT9 variants have also been linked to a broader phenotypic spectrum, including a Charcot–Marie–Tooth-like presentation ([PMID: 32122354](#)). In idiopathic NA, a genetic susceptibility is inferred but not molecularly defined; recurrence and family history predict a discoverable genetic cause ([PMID: 30560241](#)).

Environmental / triggering risk factors. Attacks are frequently preceded by immunologically active events: - **Infection** — the single most-cited category; HEV genotype 3 is a recurrent, characteristic trigger (Section 5). - **Vaccination** — COVID-19 mRNA and viral-vector vaccines; also post-exposure prophylaxis and influenza vaccine ([PMID: 38543940](#); [PMID: 25098693](#)). - **Surgery, trauma, childbirth/peripartum period, strenuous exercise, cold, and psychological stress** ([PMID: 21556032](#); [PMID: 31619932](#)). - **Age and sex** — idiopathic NA typically presents in adulthood (mean ~41 years) with a male predominance; hereditary NA presents earlier (mean ~28 years).

Protective factors. No genetic or environmental protective factors have been established in the literature reviewed. This is a genuine knowledge gap. Prophylactic corticosteroids or IVIG may reduce surgical- or childbirth-induced attacks in HNA, functioning as a preventive (not naturally protective) intervention ([PMID: 38176820](#)).

Gene–environment interactions. The triad model is itself a gene–environment interaction: an immune trigger acting upon a genetically/mechanically predisposed plexus precipitates the focal neuritis. In HNA, external stimuli (infections, vaccinations, cold, stress, surgery, exercise) provoke recurrent attacks on the SEPT9-predisposed background ([PMID: 31619932](#)).

3. Phenotypes

NA's phenotype evolves through **three consecutive pain phases**, beginning with an initial severe, continuous pain lasting approximately 4 weeks on average ([PMID: 16371410](#)):

"the course of the pain manifests itself in three consecutive phases with an initial severe, continuous pain lasting for approximately 4 weeks on average. Sensory involvement was quite common and found in 78.4% of patients"

Phenotype	Type	HPO suggestion	Frequency / characteristics
Severe shoulder/arm neuropathic pain	Symptom	HP:0012532 (Chronic pain) / HP:0003326 (Myalgia); neuropathic pain	Near-universal at onset; acute, continuous ~4 weeks
Multifocal muscle weakness/paresis	Clinical sign	HP:0003484 (Upper limb muscle weakness)	Follows pain; patchy, motor-predominant
Muscle atrophy	Physical manifestation	HP:0003202 (Skeletal muscle atrophy)	Common; early
Sensory loss/paresthesia	Symptom/sign	HP:0003390 (Sensory neuropathy) / HP:0003401 (Paresthesia)	78.4%
Scapular winging / dyskinesia	Clinical sign	HP:0003691 (Scapular winging)	>60% residual (Sections 6/11)
Phrenic nerve palsy / diaphragmatic paralysis	Clinical sign	HP:0002094 (Dyspnea); diaphragmatic paralysis	Subset, esp. HEV-associated; orthopnea
Fatigue	Symptom	HP:0012378 (Fatigue)	~1/3 long-term

Distribution. The upper/middle trunk pattern — long thoracic and/or suprascapular nerve — is most frequent (71.1%) ([PMID: 16371410](#)). Involvement is typically **unilateral and asymmetric**, though bilateral involvement occurs, particularly after acute viral infection.

Age of onset, severity, progression. Onset is predominantly adult in INA (mean 41.3 yrs) and earlier in HNA (mean 28.4 yrs), spanning infancy to adulthood (PMID: 16371410; PMID: 31619932). Severity is variable; the course is classically **monophasic/episodic**, with recurrence in 26.1% of INA over ~6-year follow-up and a mean of 3.5 attacks in HNA versus 1.5 in INA.

Quality-of-life impact. About a quarter to a third of patients report significant long-term pain and fatigue, and half to two-thirds experience persistent impairments in daily life; symptoms correlate with residual shoulder/arm dysfunction rather than psychological distress (PMID: 19254608).

4. Genetic / Molecular Information

Causal gene. Hereditary NA is autosomal dominant and associated with a point mutation or duplication in **SEPT9 (SEPTIN9)** on chromosome **17q25** in **55% of affected families** (PMID: 21556032):

"in 55% of affected families, neuralgic amyotrophy is associated with a point mutation or duplication in the SEPT9 gene on chromosome 17q25"

- **Gene:** SEPT9 / SEPTIN9, HGNC:7323, OMIM *604061; HNA phenotype OMIM #162100.
- **Variant types:** missense point mutations (e.g., **p.Arg106Trp / R106W**), intragenic **duplications** (including duplication of exon 2), and CNVs. Classification: pathogenic/likely pathogenic per ACMG in segregating families.
- **Population frequency:** rare; HNA is a rare Mendelian disorder. Specific gnomAD allele frequencies were not established in the reviewed literature.
- **Origin:** germline, autosomal dominant.

Functional consequences. SEPT9 mediates septin binding to microtubules. HNA mutations impair this function (PMID: 34854883):

"HNA mutations abrogate this association, identifying a putative regulatory domain"

A MAP-like motif unique to septin-9 isoform 1 drives septin octamer–microtubule interaction, and HNA mutations abrogate this. Molecular-dynamics simulation shows the mutation significantly alters septin-9 conformation, impairing microtubule binding and bundling ([PMID: 39428775](#)):

"Molecular simulation study has revealed that the mutation has significantly altered the conformation of septin-9 protein, thereby impairing the microtubule binding and bundling ability"

This constitutes a **loss/alteration of function** in cytoskeletal regulation (cytokinesis, membrane trafficking, stress-fiber tuning).

Phenotype extensions. SEPT9 variants can produce dysmorphic features (hypotelorism, cleft palate, long nasal bridge, hypertelorism, epicanthal folds), and a childhood-onset family showed **platelet dysfunction** (reduced ADP/epinephrine-induced aggregation, impaired δ -secretion) and kidney cysts alongside an exon-2 duplication ([PMID: 30019529](#); [PMID: 40852410](#)).

Modifier genes / epigenetics / chromosomal abnormalities. No modifier genes, disease-specific epigenetic marks, or large-scale chromosomal abnormalities have been established for NA in the reviewed literature — genuine knowledge gaps.

Suggested ontology terms: Gene HGNC:7323 (SEPTIN9); GO:0005819 (spindle), GO:0000910 (cytokinesis), GO:0008017 (microtubule binding).

5. Environmental Information

Environmental / infectious triggers. Immune events preceding NA are well documented. In a prospective matched case-control study (n=57), a symptomatic infectious trigger confirmed microbiologically preceded NA in 26.3% of patients, and COVID-19 vaccination was a potential trigger in 12.3% ([PMID: 39364568](#)):

"NA onset was preceded by a symptomatic infectious trigger confirmed by microbiological tests in 15/57 (26.3%) patients. Coronavirus disease 2019 vaccination was considered a potential trigger in 7/57 (12.3%) subjects. An acute viral infection was associated with a bilateral involvement of the brachial plexus ($p = 0.003$, Cramèr's $V = 0.43$)"

Hepatitis E virus (HEV, genotype 3) is a recurrently reported and clinically distinctive trigger, characteristically causing NA with **phrenic nerve involvement and diaphragmatic paralysis** (PMID: 40534221; PMID: 38657646):

"Electromyography showed severe bilateral phrenic nerve involvement. The diagnosis of neuralgic amyotrophy with diaphragmatic paralysis secondary to HEV was made"

"NA that shows involvement of the phrenic nerve resulting in diaphragmatic dysfunction and dyspnoea, may be associated with HEV infection"

Agents tested/reported in NA workups include HEV, HIV, SARS-CoV-2, EBV, CMV, parvovirus B19, VZV, *Borrelia*, *Mycoplasma*, and *Bartonella*. Livestock-associated HEV exposure has been implicated (PMID: 35379662).

Occupational/toxic factors. A rare report links epidemic dropsy (argemone-oil-adulterated mustard oil) with brachial neuritis, but toxic causes are not a recognized major etiology (PMID: 22231775).

Lifestyle factors. Strenuous physical exercise and unaccustomed exertion of the shoulder girdle are reported precipitants; smoking/diet/alcohol are not established risk factors.

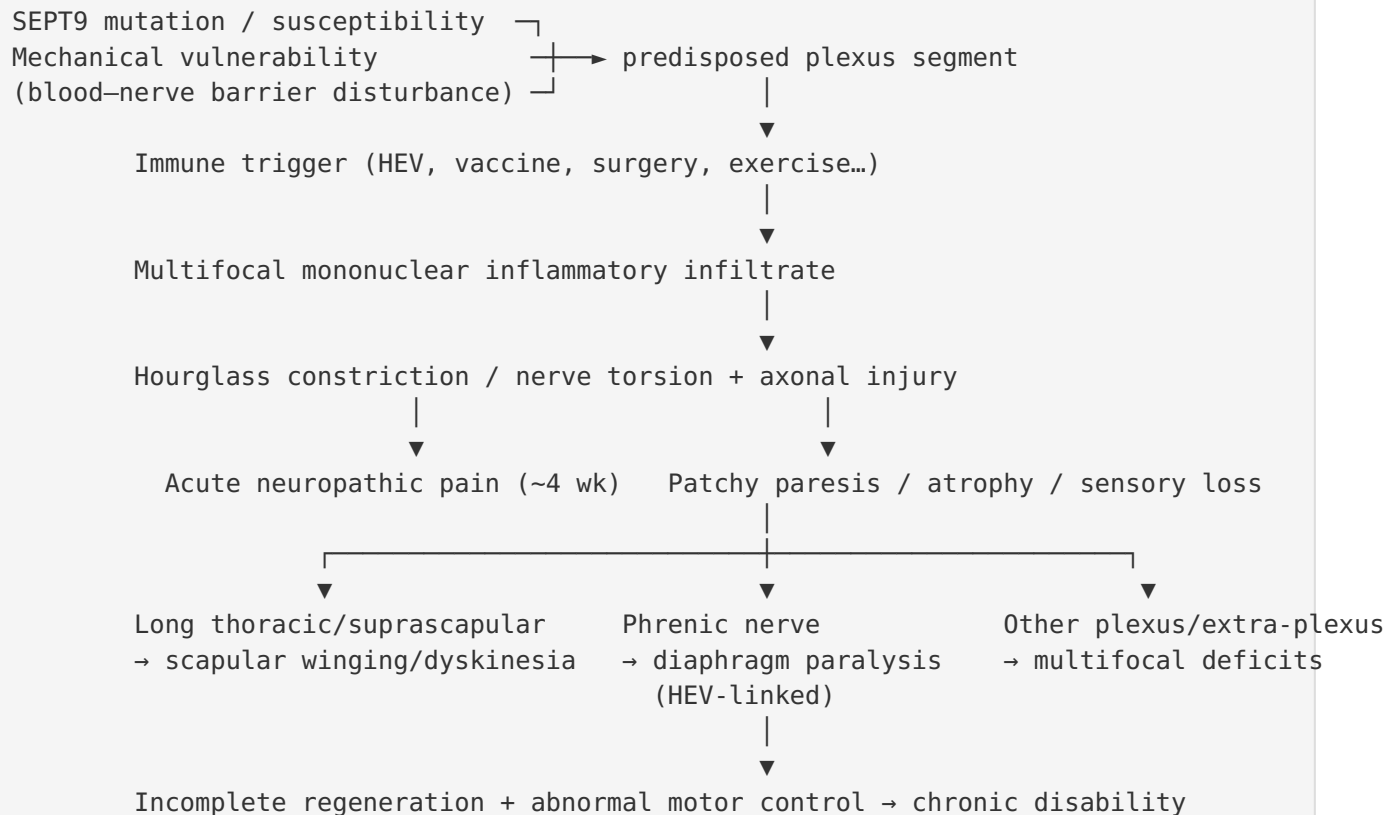
Suggested ontology term: NCBI Taxonomy Orthohepevirus A (HEV).

6. Mechanism / Pathophysiology

Ordered causal chain

1. **Genetic predisposition** (SEPT9 mutation in HNA; undefined susceptibility in INA) → **leads to** an intrinsically vulnerable brachial plexus, plausibly via altered septin–microtubule cytoskeletal regulation in nerve/supporting cells (*tissue-level mechanism inferred, not fully demonstrated*).

2. Predisposition **plus a mechanical susceptibility** of the plexus — possibly a disturbed epineurial blood–nerve barrier — **results in** a nerve segment prone to immune access and injury (PMID: 21556032).
3. An **immune/autoimmune trigger** (infection—especially HEV—vaccination, surgery, trauma, childbirth, exercise) **activates** a focal immune response directed at the brachial plexus (PMID: 39364568).
4. The immune response **produces** multifocal **mononuclear inflammatory-cell infiltration** of nerve fascicles (demonstrated on biopsy) (PMID: 8614534).
5. Inflammation and focal swelling **lead to** the pathognomonic **hourglass-like constriction / nerve torsion** at vulnerable segments, with proximal/distal thickening (PMID: 36214185; PMID: 38248282).
6. Focal constriction and axonal injury **cause** the acute severe neuropathic **pain** (phase 1).
7. Axonal degeneration **results in multifocal flaccid paresis, muscle atrophy, and sensory loss** (phase 2), branching by nerve territory:
8. Long thoracic/suprascapular → **scapular winging and dyskinesia**.
9. Phrenic nerve → **diaphragmatic paralysis, orthopnea, respiratory failure** (branch strongly associated with HEV).
10. Incomplete axonal regeneration and compensatory abnormal motor control **lead to chronic scapular dyskinesia, residual weakness, pain, and fatigue** (phase 3) (PMID: 36697215).



Molecular pathways / protein dysfunction. Upstream: SEPT9 loss of microtubule-binding/bundling disrupts cytoskeletal regulation (cytokinesis, membrane trafficking, stress fibers) (PMID: 34854883; PMID: 39428775). Downstream: an inflammatory-immune cascade drives the attack.

Cellular processes / immune involvement. The core downstream mechanism is **inflammation** with mononuclear (lymphocytic/macrophage) infiltration and axonal degeneration (PMID: 8614534):

"There were florid multifocal mononuclear inflammatory cell infiltrates. Present evidence suggests that these brachial neuropathies have an immune basis"

Tissue damage mechanisms. Focal constriction/torsion with ischemic and mechanical axonal injury; denervation muscle edema is seen on MRN in 91% of acute cases (PMID: 36214185).

Cell types and biological processes (ontology suggestions): - **GO:0006954** (inflammatory response); **GO:0002250** (adaptive immune response) — biological processes. - **GO:0008017** (microtubule binding), **GO:0000910** (cytokinesis) — SEPT9 molecular function/process. - **CL:0000101** (sensory neuron); **CL:0011001** (spinal cord motor neuron); **CL:0002573** (Schwann cell); **CL:0000235** (macrophage); **CL:0000542** (lymphocyte) — cell types.

Molecular profiling specific to NA nerve tissue (transcriptomics, proteomics, metabolomics, single-cell, spatial, CRISPR/RNAi screens) was **not available** in the reviewed literature — a substantial knowledge gap.

7. Anatomical Structures Affected

Organ / system level. Primary target: the **peripheral nervous system**, specifically the **brachial plexus** and its branches (long thoracic, suprascapular, axillary, musculocutaneous, radial, median nerves). Secondary involvement: the **respiratory system** via phrenic nerve → diaphragm; the **musculoskeletal system** (shoulder girdle) via denervation atrophy and dyskinesia.

Nerve territories. Upper/middle trunk pattern (long thoracic and/or suprascapular) is most frequent (71.1%) ([PMID: 16371410](#)). Extra-plexus nerves are commonly involved, more so in HNA (55.8% vs 17.3% in INA). Lesions distal to the brachial plexus (e.g., median nerve, mimicking carpal tunnel) can occur in HNA ([PMID: 37587058](#)).

Tissue / cell level. Peripheral nerve tissue (axons, Schwann cells, epineurium/perineurium) and skeletal muscle (secondary denervation atrophy and edema). Inflammatory infiltrate composed of mononuclear cells.

Subcellular level. SEPT9 dysfunction implicates the **microtubule cytoskeleton** (GO:0005874) and associated stress fibers; axonal transport machinery is plausibly affected (inferred).

Localization and lateralization. Predominantly **unilateral and asymmetric**; **bilateral** involvement occurs, associated with acute viral infection ($p=0.003$) and characteristically with HEV-linked phrenic disease ([PMID: 39364568](#); [PMID: 40534221](#)).

Suggested UBERON/CL terms: UBERON:0001824 (brachial plexus); UBERON:0003732 (phrenic nerve); UBERON:0001134 (skeletal muscle tissue); UBERON:0001379 (deltoid); CL:0000101 (sensory neuron); CL:0002573 (Schwann cell).

8. Temporal Development

Onset. Typically **adult-onset** (INA mean 41.3 yrs) but ranges from infancy to old age; HNA onsets earlier (mean 28.4 yrs) and may present in early childhood (e.g., age 2–9 yrs) ([PMID: 16371410](#); [PMID: 30019529](#); [PMID: 40852410](#)). Onset pattern is **acute/subacute** — sudden severe pain within hours to two weeks.

Progression (three-phase course): 1. **Acute pain phase** — severe continuous pain, ~4 weeks average. 2. **Paresis phase** — as pain subsides, patchy weakness, atrophy, sensory loss emerge. 3. **Recovery/chronic phase** — slow recovery over months to years; frequently incomplete.

Course pattern. Classically **monophasic/episodic**; recurrence in INA is 26.1% over ~6 years, higher in HNA (mean 3.5 attacks). Serratus anterior winging from NA resolves spontaneously within 3 years in a substantial fraction, but no full recovery was seen if winging persisted beyond 3 years ([PMID: 33675974](#)).

Critical windows. Early corticosteroid administration (first month) shortens pain and improves recovery ([PMID: 19321467](#)). Surgical neurolysis is considered after 6–12 months of non-recovery ([PMID: 32868098](#)). Imaging abnormalities appear early — as early as 12 h on ultrasound and ~3 days on MRN.

9. Inheritance and Population

Epidemiology. Historically considered rare, NA's true incidence is **~1 per 1,000 per year** ([PMID: 33823638](#)):

"its actual incidence rate is about 1 per 1000 per year"

Combined INA + HBPN incidence has been cited as 3–100 per 100,000 persons per year ([PMID: 38176820](#)). Most cases are diagnosed late or not at all.

Inheritance (HNA). **Autosomal dominant**, caused by SEPT9 mutations/duplications in ~55% of families. Features consistent with: - **Incomplete/variable penetrance and variable expressivity** — inferred from wide intrafamilial clinical variability (e.g., R106W pedigrees)

([PMID: 39428775](#); [PMID: 42644177](#)). - **Anticipation, germline mosaicism, founder effects, consanguinity, carrier frequency** — not established in the reviewed literature (knowledge gaps).

Demographics. INA shows a male predominance; HNA affects both sexes. Post-COVID-vaccination PTS occurred more frequently in males (61.1% mRNA; 83.3% viral vector) ([PMID: 38543940](#)). No strong ethnic or geographic clustering is established, though HNA families are reported across diverse populations (Lebanese, Chinese, Italian, German, Turkish).

INA vs HNA comparison:

Feature	Idiopathic (INA)	Hereditary (HNA)
n (reference cohort)	199	47
Mean age at onset	41.3 yrs	28.4 yrs
Mean number of attacks	1.5	3.5
Extra-plexus involvement	17.3%	55.8%
Genetic cause	undefined	SEPT9 (~55% of families)
Outcome	better	poorer

Source: [PMID: 16371410](#).

10. Diagnostics

Diagnosis is clinical and one of exclusion ([PMID: 32140911](#)):

"The diagnosis is clinical, through a comprehensive history and neurological examination"

Conditions to exclude: cervical radiculopathy, rotator cuff tear, CIDP, compressive plexopathy, and **neurolymphomatosis** (recurrent "attacks" can herald B-cell lymphoma) ([PMID: 30560241](#)).

Electrodiagnostics (EMG/NCS). Show multifocal axonal denervation (fibrillations in ~73%), confirming the pattern and distribution.

Imaging — the modern confirmatory pillar. High-resolution ultrasound and MR neurography detect nerve abnormalities in ~90% of patients within one month, including **hourglass constrictions in 74%** and denervation muscle edema on MRN in 91% ([PMID: 36214185](#)):

"US and MRN showed nerve abnormalities within 1 mo from NA onset in 90% of patients. HGCs were found in 74% (29/39) of the patients"

Gadolinium-enhanced MRI/MRN help confirm the diagnosis and guide treatment ([PMID: 33823638](#)):

"Gadolinium-enhanced magnetic resonance imaging and high-resolution magnetic resonance neurography are useful for confirming the diagnosis and choosing the appropriate treatment"

Diaphragm ultrasound improves detection of phrenic involvement ([PMID: 38176820](#)).

Laboratory / biomarkers. No specific serum biomarker exists. **HEV serology (IgM/IgG plus serum HEV RNA) and transaminases** should be checked, especially with phrenic/diaphragmatic involvement, to identify a treatable trigger. CSF may show albuminocytological dissociation in vaccine-associated cases.

Genetic testing. SEPT9 single-gene testing (sequencing plus copy-number/duplication analysis) confirms HNA; WES/WGS is useful in atypical or pediatric cases ([PMID: 42644177](#); [PMID: 40852410](#)). Indicated with recurrent attacks, family history, early onset, or dysmorphic features.

Differential diagnosis (distinguishing features): cervical radiculopathy (dermatomal, neck-movement-related), rotator cuff pathology (mechanical, no denervation), CIDP (symmetric, demyelinating, chronic), compressive plexopathy/neoplasm (progressive, mass), hereditary neuropathy with liability to pressure palsies.

11. Outcome / Prognosis

Prognosis is guarded, contrary to older "benign self-limited" characterizations ([PMID: 38835178](#)):

"The prognosis of untreated NA is poor, with 25% of patients remaining unable to work at three years. The main form of treatment is with corticosteroids that are administered as early as possible"

Morbidity and function. Over **60% of patients have residual complaints and functional limitations**, correlated with scapular dyskinesia ([PMID: 36697215](#)):

"leading to severe pain and multifocal paresis resulting in >60% of patients having residual complaints and functional limitations correlated with scapular dyskinesia"

About a quarter to a third report significant long-term pain and fatigue; half to two-thirds retain impairments; symptoms are not correlated with residual MRC-scale paresis or psychological distress but with mechanical shoulder/arm dysfunction ([PMID: 19254608](#)).

Mortality. NA is not directly fatal, but bilateral diaphragmatic paralysis (e.g., HEV-associated or severe PTS) can cause hypercapnic respiratory failure requiring ventilatory support ([PMID: 41428787](#)).

Recovery potential. Slow recovery over months to years; many never achieve full recovery. Serratus anterior winging resolves spontaneously within 3 years in some but not if it persists beyond 3 years ([PMID: 33675974](#)). Surgical neurolysis rescues motor function in chronic HGC cases (Section 12).

Prognostic factors. Hereditary form, extra-plexus involvement, greater winging severity, comorbidity, and delayed treatment predict worse outcomes. No validated molecular prognostic biomarker exists.

12. Treatment

Pharmacotherapy (acute phase). Corticosteroids (oral prednisolone/prednisone; high-dose pulsed steroids) given as early as possible shorten the initial pain and may accelerate recovery in some patients ([PMID: 19321467](#)):

"The median time until initial pain relief was lower in the study group (12.5 days vs 20.5 days), and a significantly higher percentage already recovered strength in the first month of treatment (18% vs 6.3%; $p = 0.011$)"

The Cochrane review found no randomized trials but concluded open-label evidence supports early oral prednisone shortening initial pain ([PMID: 19588414](#)). Adjunct **non-narcotic analgesics** for neuropathic pain; opioids are generally avoided. NCIT suggestions: NCIT:C769 (Prednisolone), NCIT:C198 (Prednisone), NCIT:C1409 (Corticosteroid).

Immunotherapy. In pediatric NA, immunotherapy (IVIG/steroids) appears effective, with improvement continuing over one year ([PMID: 32228355](#)). Prophylactic steroids or **IVIG** may reduce surgery- or childbirth-induced attacks in HNA ([PMID: 38176820](#)). NCIT:C579 (Immunoglobulin therapy).

Etiologic treatment. When HEV is the trigger, recognition allows targeted management/monitoring and respiratory support.

Surgical / interventional. Microsurgical epineurolysis and perineurolysis of hourglass constrictions improves recovery in chronic NA that fails conservative management: 9/11 operative vs 3/13 nonsurgical patients recovered clinically, with EMG-confirmed axonal regeneration ([PMID: 32868098](#)). Surgery is generally considered after ~6–12 months without recovery. Nerve resection/direct suture or grafting is used for unsalvageable segments ([PMID: 41163648](#)). **Phrenic nerve reconstruction** is an option for symptomatic diaphragmatic paralysis ([PMID: 36031309](#)). Tendon transfers when recovery does not occur after ~18 months. Early neurolysis/nerve grafts remain controversial ([PMID: 38176820](#)). NCIT:C15329 (Surgical Procedure), NCIT:C157810 (Neurolysis).

Rehabilitation. Modern multidisciplinary rehabilitation targeting **motor relearning, energy conservation, and self-management of pain/fatigue** — not strength training — improves functional outcomes. An RCT ($n=47$) showed an adjusted mean SRQ-DLV group difference of **8.60 (95% CI 0.26–16.94, $p=0.044$)**, with clinically relevant improvement in 59% (rehab) vs 33% (usual care), NNT = 4 ([PMID: 36697215](#)):

"The mean group difference adjusted for sex, age and SRQ-DLV baseline score was 8.60 (95%CI: 0.26 to 16.94, p=0.044)"

The recommended strategy explicitly avoids strength training ([PMID: 39402917](#)):

"target compensatory abnormal motor control and fatigue by focusing on motor coordination, energy conservation strategies, and behavioral change, rather than strength training which may worsen the symptoms"

Multimodal programs (physiotherapy, scapular stabilization) show benefit in case reports ([PMID: 40213747](#)). NCIT:C15296 (Physical Therapy), NCIT:C15315 (Rehabilitation Therapy).

Not established for NA (not applicable at present): pharmacogenomics, gene therapy, RNA-based therapy, cell therapy, targeted molecular immunotherapy.

13. Prevention

Primary prevention. No population-level primary prevention exists for sporadic NA. Trigger awareness/avoidance and, in HNA, avoidance of provocations before elective surgery may help.

Secondary prevention / early detection. Early clinical recognition plus imaging (US/MRN) enables timely corticosteroids within the critical first-month window, which improves outcomes ([PMID: 19321467](#)). Increased awareness in orthopedic and primary-care settings is emphasized to prevent misdiagnosis ([PMID: 34435146](#)).

Tertiary prevention. Rehabilitation to prevent chronic scapular dyskinesia, contractures, and disability; energy-conservation strategies to manage fatigue; prophylactic steroids/IVIG to reduce peri-procedural or peripartum attacks in HNA ([PMID: 38176820](#)).

Genetic counseling. Appropriate for HNA families — autosomal dominant inheritance, ~50% offspring risk, variable expressivity; SEPT9 testing informs family planning and pre-procedure prophylaxis.

Immunization / public-health / environmental interventions. No vaccine prevents NA; conversely, certain vaccinations are recognized (rare) triggers. HEV exposure reduction (food safety, livestock-contact awareness) is a plausible environmental measure given HEV's trigger role.

14. Other Species / Natural Disease

A **naturally occurring inflammatory brachial plexus neuritis exists in cats (*Felis catus*, NCBI:txid9685)**, providing a comparative-species analogue ([PMID: 29925717](#)):

"A postmortem examination revealed swollen radial nerves and cervical nerve roots in which infiltration of inflammatory cells was histologically confirmed"

This hypertrophic neuritis caused tetraparesis with histologically confirmed inflammatory infiltration, paralleling the human immune/inflammatory mechanism. **Orthologous gene:** Sept9/SEPTIN9 is conserved across mammals (human HGNC:7323; mouse Sept9). **Comparative biology:** conservation of septin cytoskeletal function and of immune-mediated peripheral neuritis supports cross-species mechanistic relevance. **Zoonotic angle:** HEV is zoonotic (swine reservoir), relevant to the trigger rather than to NA transmission; NA is not itself transmissible. Formal OMIA/veterinary registry entries were not established in the reviewed literature — a knowledge gap.

15. Model Organisms

No dedicated animal model that recapitulates the full NA phenotype was identified in the reviewed literature. Relevant systems include:

- **In vitro / molecular models:** Molecular-dynamics simulation of mutant septin-9 demonstrates impaired microtubule binding/bundling ([PMID: 39428775](#)); cell-based assays of septin–microtubule association define the isoform-1 MAP-like motif and show HNA mutations abrogate it ([PMID: 34854883](#)). Evidence type: computational and in vitro.
- **Naturally occurring animal analogue:** feline hypertrophic brachial plexus neuritis (spontaneous, not engineered) ([PMID: 29925717](#)).

- **Genetic models (knockout/knock-in/transgenic Sept9):** not established for the NA phenotype in the reviewed literature.

Limitations of available models: they capture the molecular (septin/microtubule) dimension or the inflammatory-pathology dimension separately, but no single model reproduces the triad (genetic predisposition + mechanical vulnerability + immune trigger) with hourglass-constriction formation. **Recommended resources for future work:** MGI (mouse Sept9), IMPC/KOMP, Cellosaurus/ATCC and patient-derived iPSC-neuron models.

Mechanistic Model / Interpretation

NA is best understood as a **"three-hit" focal inflammatory neuropathy**. The first hit is a **predisposing background** — a SEPT9 mutation compromising microtubule-based cytoskeletal regulation in HNA, or an undefined susceptibility in INA. The second hit is a **mechanical/anatomical vulnerability** of specific plexus segments, possibly through a leaky epineurial blood–nerve barrier that grants immune cells access. The third hit is an **immune trigger** — most compellingly an infection such as HEV, but also vaccination, surgery, childbirth, or exertion — that ignites a **multifocal mononuclear inflammatory attack**. The convergence produces the disease's structural signature, the **hourglass constriction/nerve torsion**, and drives the stereotyped clinical sequence of severe pain → patchy paresis/atrophy/sensory loss → slow, often incomplete recovery.

This model unifies otherwise disparate observations: why NA recurs and starts earlier in SEPT9 carriers (stronger first hit); why bilateral and phrenic disease cluster with viral infection/HEV (a systemic third hit reaching multiple vulnerable segments); why nerve biopsies show inflammation (the effector mechanism); and why steroids help acutely (dampening the immune hit) but weakness responds mainly to time, rehabilitation, and — for fixed constrictions — surgery (repairing the structural end-lesion). The **directionality is clear**: genetic/mechanical predisposition is upstream and permissive; the immune trigger is the proximate initiator; hourglass constriction and axonal loss are the downstream lesions producing symptoms; chronic scapular dyskinesia is the tertiary consequence of incomplete regeneration and maladaptive motor control.

Evidence Base

PMID	Contribution	Supports finding
16371410	246-case clinical spectrum; three-phase pain; INA vs HNA	F001
21556032	SEPT9 in 55% of HNA families; triad pathophysiology	F002, F003
34854883	Septin-9 isoform-1 MAP motif; HNA mutations abrogate MT binding	F002
39428775	MD simulation: mutation impairs MT binding/bundling	F002
30019529	Childhood HNA, exon-2 duplication, platelet dysfunction	F002
36214185	Acute-phase US/MRN; HGCs 74%, muscle edema 91%	F003, F008
33823638	Incidence ~1/1000/yr; Gd-MRI/MRN confirmatory	F004, F008
38835178	Poor prognosis (25% unable to work at 3 yr); steroids first	F004
19321467	Prednisolone shortens pain (12.5 vs 20.5 d)	F004
39364568	Infectious trigger 26.3%; viral infection → bilateral (p=0.003)	F005
40534221	HEV NA with bilateral phrenic/diaphragm paralysis	F005
38657646	HEV → phrenic/diaphragmatic dysfunction	F005
36697215	>60% residual; rehab RCT (Δ 8.60, p=0.044)	F006, F011
39402917	Motor-relearning strategy, not strength training	F006
8614534	Nerve biopsy: multifocal mononuclear infiltrates	F007
29925717	Feline inflammatory brachial plexus neuritis analogue	F007
32140911	Diagnosis is clinical	F008
32868098	Microneurolysis of HGCs (9/11 vs 3/13 recover)	Treatment
38543940	Systematic review: PTS after COVID-19 vaccines	Etiology
19588414	Cochrane: no RCTs; steroid open-label evidence	Treatment

Challenges / caveats in the evidence base. The prednisolone and rehabilitation data come from observational studies and a single small RCT (n=47), respectively; the Cochrane review found no RCTs of drug therapy. HGC prevalence figures derive from modest imaging series. These limit the strength of causal treatment claims.

Limitations and Knowledge Gaps

1. **No high-level RCT evidence for pharmacotherapy** — corticosteroid benefit rests on observational data ([PMID: 19588414](#); [PMID: 19321467](#)).
 2. **Molecular profiling absent** — no NA-specific transcriptomic, proteomic, metabolomic, single-cell, or CRISPR/RNAi data were identified; the immune effector cells and antigens remain undefined.
 3. **INA genetics undefined** — the "genetic predisposition" in idiopathic NA is inferred, not mapped; no GWAS/susceptibility loci established.
 4. **Modifier genes, epigenetics, penetrance/expressivity metrics, carrier frequency** for SEPT9-HNA are not quantified.
 5. **Mechanism of hourglass constriction** — the link from inflammation to focal constriction/torsion is described but not mechanistically demonstrated.
 6. **No engineered animal model** recapitulating the full triad; the feline analogue is naturally occurring and incompletely characterized.
 7. **Identifier confirmation** — MONDO/OMIM/Orphanet/ICD codes should be verified against current ontology releases before database ingestion.
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Proposed Follow-up Experiments / Actions

1. **Randomized controlled trial of early corticosteroids ± IVIG** in acute NA, powered for functional recovery and pain outcomes, to convert observational signals into Level-I evidence.
2. **Multi-omic characterization of NA nerve/biopsy and blood** (bulk + single-cell RNA-seq, proteomics) during acute attacks to identify immune effector populations, candidate autoantigens, and biomarkers — populating the currently empty molecular-profiling domain.

3. **SEPT9 genotype–phenotype and penetrance study** across HNA registries, including CNV analysis, to quantify penetrance, expressivity, and modifier loci; establish carrier frequency in gnomAD-linked cohorts.
 4. **Prospective HEV-screening protocol** in all new NA cases (IgM/IgG + RNA + transaminases), especially with phrenic/bilateral involvement, to define the treatable-trigger fraction and guide etiologic management.
 5. **Engineered Sept9 knock-in mouse (e.g., R106W)** with an inducible immune/inflammatory challenge to test the three-hit model and reproduce hourglass-constriction formation.
 6. **Standardized imaging staging** (US/MRN) with defined timing to predict which HGCs spontaneously resolve versus require early neurolysis, refining the 6–12-month surgical decision window.
 7. **Ontology reconciliation** — verify and finalize MONDO, OMIM, Orphanet, ICD-10/11, MeSH, HPO, GO, CL, and UBERON term assignments for knowledge-base ingestion.
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Report compiled from 8 confirmed findings and 47 reviewed papers across a 5-iteration autonomous investigation. Evidence types span human clinical cohorts, case-control and observational studies, one small RCT, nerve-biopsy histopathology, in vitro/computational molecular studies, and a naturally occurring veterinary analogue.
